

Shubham Sarkar

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PROFESSIONAL SUMMARY

Detail-oriented Machine Learning Engineer with hands-on experience in building data pipelines, deploying models, and optimizing infrastructure for ML workloads. Proven track record in enhancing model performance and reliability through innovative solutions and effective data management strategies.

WORK EXPERIENCE

Machine Learning Engineer

Jan 2026

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- Developed Python scripts for ingesting, cleaning, and transforming metadata datasets, enhancing data quality for ML applications.
- Constructed preprocessing and augmentation pipelines for audio and image data, improving model readiness and performance.
- Evaluated AI agent responses, identifying failure points and enhancing model accuracy by 20% through targeted improvements.

Deep Learning Research Assistant

May 2025 – September 2025

Jadavpur University CMATER Lab

- Engineered a self-attention mechanism within a pre-trained VGG16 model, significantly improving feature extraction for lung cancer detection.
- Achieved 99.54% peak accuracy with a hybrid deep learning architecture, optimizing for edge-device deployment with minimal resource usage.
- Implemented data versioning and lineage tracking for model training datasets, ensuring reproducibility and compliance.

PROJECTS

TailorCV.ai

- Developed an AI web application that optimizes resumes to job descriptions using LLMs and NLP pipelines, increasing resume relevance by 80%.
- Designed a robust backend using Python and FastAPI, deployed on AWS ECS, ensuring scalability and performance.
- Achieved over 50 users within the first week of launch, demonstrating strong market demand and user engagement.

YouTube Sentiment Analysis

- Created an end-to-end sentiment analysis pipeline for YouTube comments, processing over 10,000 user inputs and enhancing classification accuracy.
- Utilized MLflow and DVC for tracking model experiments, facilitating reproducible training and systematic model evaluation.
- Deployed the solution on AWS, utilizing Docker for containerization and Flask for RESTful API integration.

Smart Product Pricing

- Developed a hybrid NLP and CV pipeline to analyze 150,000 image and text datasets for price prediction, integrating transformer-based encoders.
- Implemented comprehensive data preprocessing techniques, including text cleaning and image feature extraction, to enhance model performance.
- Achieved a rank of 142 out of 50,000 participants in a competitive data science challenge, showcasing model effectiveness.

TECHNICAL SKILLS

Machine Learning

Deep Learning

Natural Language Processing (NLP)

Computer Vision

Data Pipeline Development

AWS (S3, EC2, SageMaker)

Docker

MLflow

Data Versioning

Python

CI/CD

EDUCATION

Bachelor of Technology

Jadavpur University, Kolkata

Higher Secondary

Hariyana Vidya Mandir, Kolkata

EXTRACURRICULAR ACTIVITIES

Core Member

Entrepreneurship Cell, Jadavpur University

- ▶ Organized national-level events such as E-Summit 2025 and Hult Prize 2025, attracting over 5,000 registrations and 1,000 attendees.
- ▶ Contributed to the establishment of an Incubation Center at Jadavpur University under the Institution's Innovation Council (IIC).